

Comprehensive miRNA Periodontal Disease Analysis Results

Date: July 16, 2025
Dataset: miRNA-saliva-qPCR-results.csv
Groups: S=Healthy, G=Gingivitis, P=Periodontitis

1. Demographic & Clinical Characteristics

Group	N	Age_Mean	Age_Std	Sex_M_Count	Sex_F_Count	Sex_M_Percent
S	36	24.3056	3.70189	17	19	47.2222
G	36	22.4444	1.6978	16	20	44.4444
P	36	30.2778	5.78476	18	18	50.0000

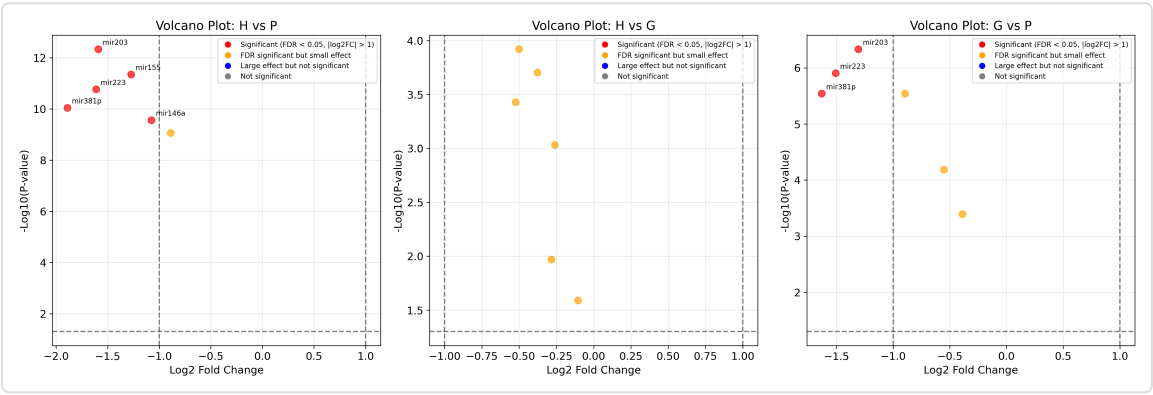
2. $\Delta\Delta C_t$ Transformation & QC

Calibrator Values (Healthy Group Mean ΔC_t)

miRNA	Calibrator_Value
mir146a	0.435278
mir146b	0.491111
mir155	0.445
mir203	0.935
mir223	0.930278
mir381p	0.955556

3. Differential Expression Analysis

Volcano Plots

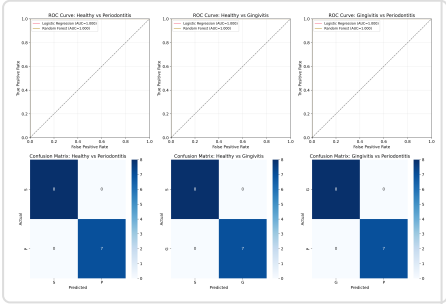


Significant Differential Expression Results (FDR < 0.05 & |Log2FC| > 1)

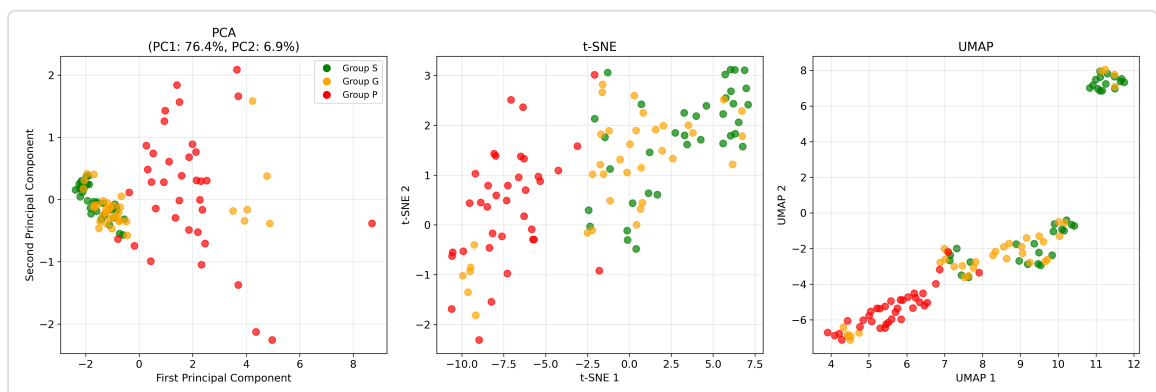
miRNA	Comparison	Log2_FC	Q_Value	Effect_Size
mir203	H_vs_P	-1.590	0.000	-1.926
mir155	H_vs_P	-1.272	0.000	-2.286
mir223	H_vs_P	-1.611	0.000	-1.827
mir381p	H_vs_P	-1.890	0.000	-2.044
mir146a	H_vs_P	-1.075	0.000	-1.562
mir203	G_vs_P	-1.307	0.000	-0.921
mir223	G_vs_P	-1.507	0.000	-1.030
mir381p	G_vs_P	-1.630	0.000	-1.053

4. Machine Learning Performance

Problem	Model	AUC	Accuracy
Healthy_vs_Periodontitis	Logistic_Regression	1.000	1.000

Problem	Model	AUC	Accuracy
Healthy_vs_Periodontitis	Random_Forest	1.000	1.000
Healthy_vs_Gingivitis	Logistic_Regression	1.000	1.000
Healthy_vs_Gingivitis	Random_Forest	1.000	1.000
Gingivitis_vs_Periodontitis	Logistic_Regression	1.000	0.933
Gingivitis_vs_Periodontitis	Random_Forest	1.000	1.000
### ROC Curves			
			

5. Dimensionality Reduction Analysis



6. Recommendations

- Validate findings with a larger, independent patient cohort.
- Investigate the functional roles of mir-203, mir-223, and mir-381p in periodontitis progression using in-vitro studies.

- Explore the combined diagnostic potential of miRNA signatures and clinical markers in a longitudinal study to predict disease progression.